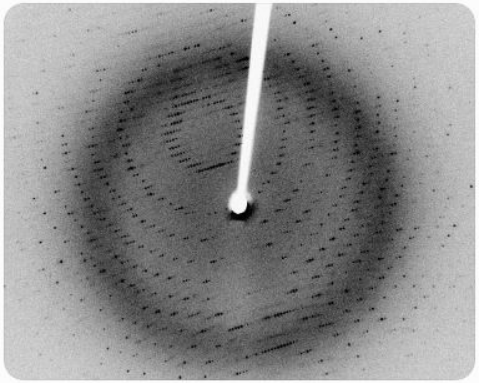


X-ray crystallography

Detector image containing Bragg reflections from a crystal lattice.

Diffraction images

Spot finding

Indexing and integration

Scaling and merging

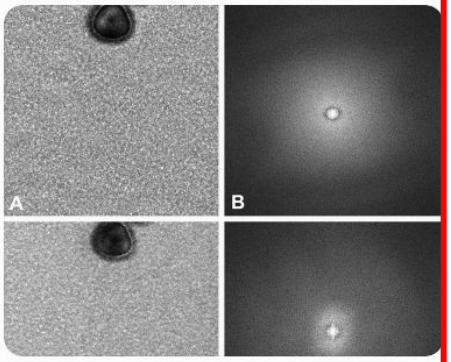
Phasing

Electron-density map

Atomic model building

Reciprocal-space refinement

Validation



Cryo-electron microscopy

Representative particle micrographs and their Fourier transforms; not the PP2A dataset.

Cryosparc, TOPAZ

Relion, EMReady,COOT

Electron movies

Motion and CTF correction

Particle picking

2D classification

3D reconstruction

3D classification and refinement

3D density map

Atomic model building

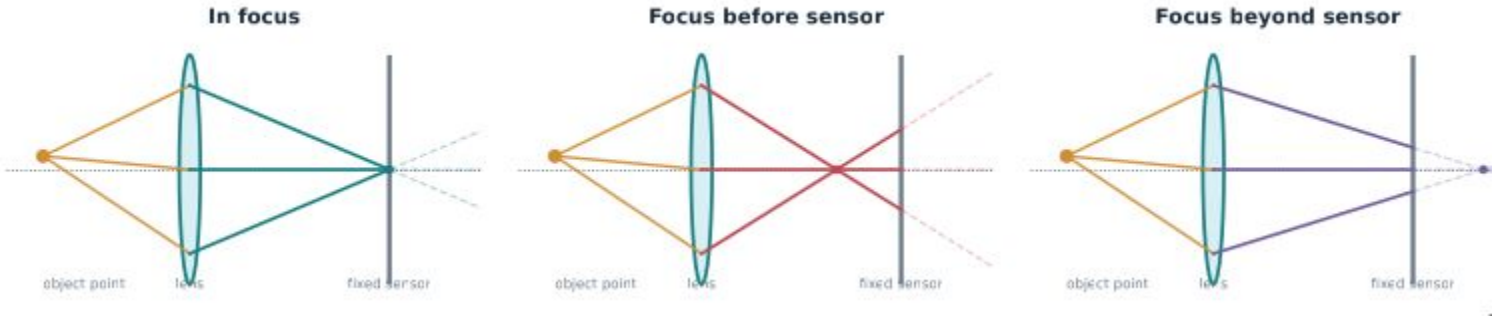
Real-space refinement

Validation

What is cryoEM downstream analysis? How does it differ from x ray crystallography?

Background Information/Refresher

What does it mean to defocus a lens

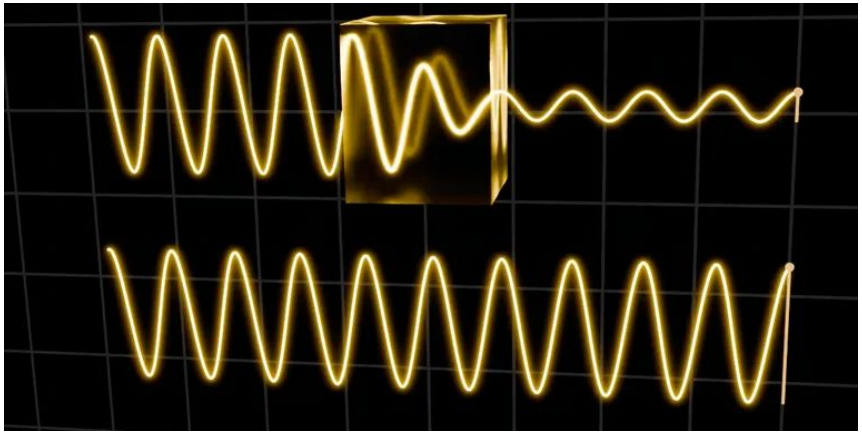


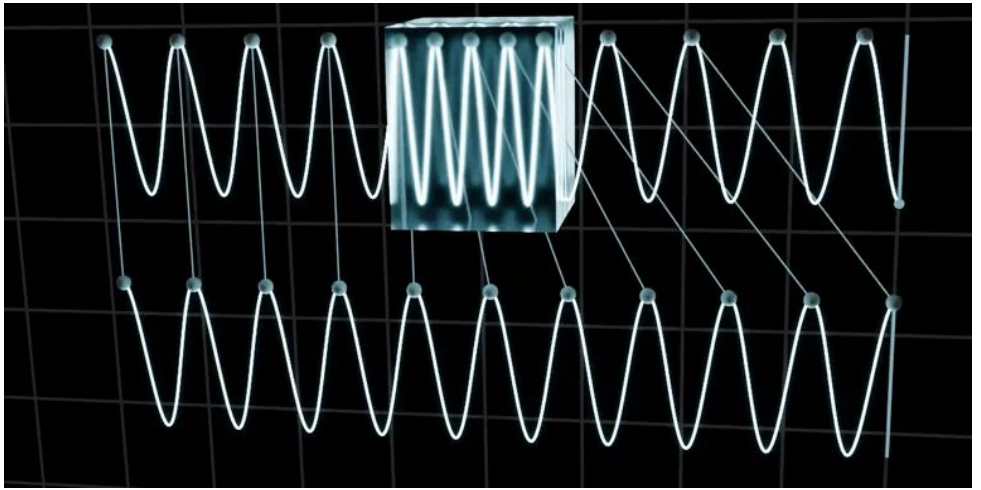
An ordinary thin lens satisfies

$$\frac{1}{f_L} = \frac{1}{s} + \frac{1}{s'}, \qquad s' = \frac{s f_L}{s - f_L}, \tag{1}$$

where f_L is the focal length, s the object-to-lens distance, and s' the image distance. If the sensor is placed at s', it records the focused image. Changing f_L while holding the object and sensor fixed shifts the best image plane.

What happens to electron beam as it passes through protein?

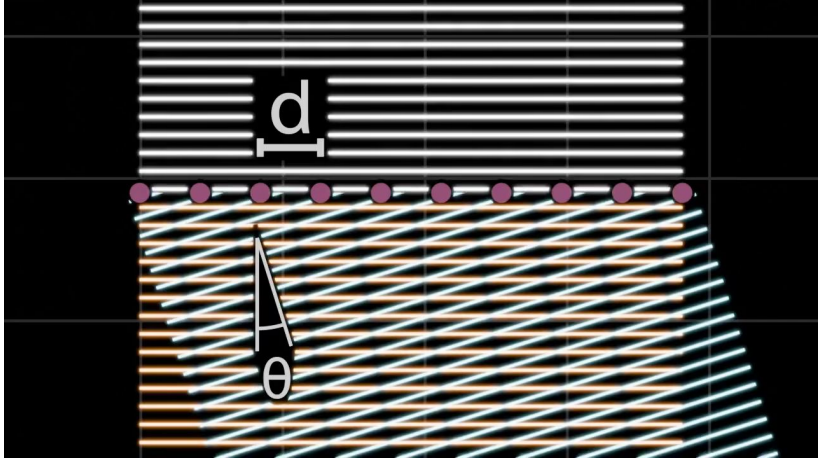


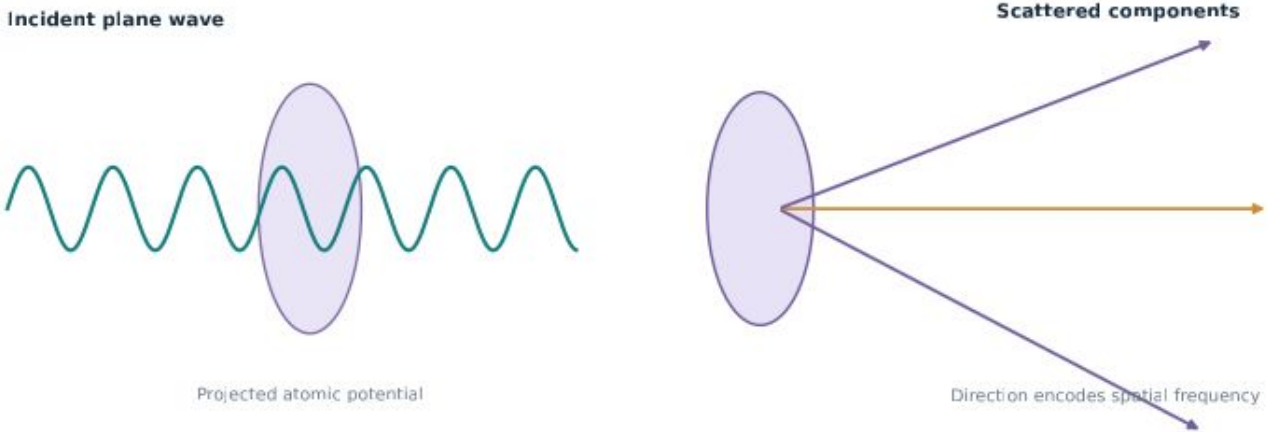


Effect 1: Amplitude decrease and phase shift(mainly)

Bragg law:
Where theta is scatter angle and d is the spatial resolution(spacing) of the protein.

So spatial frequency(mentioned later) is just 1/d





Effect 2: scattering due to diffraction

Key assumption used in cryoEM imaging: Weak phase object approximation
Impliciation: scattered wave lengths are always 90 degree out of phase compared to unscattered

$$\Psi_{\text{in}}(\mathbf{r}, z) = e^{ikz}, \qquad k = \frac{2\pi}{\lambda}$$

$$\Psi_{\text{exit}}(x, y) = e^{i\phi(x,y)}$$

Under the WPOA, $|\phi| \ll 1$, so Taylor expansion gives

$$e^{i\phi} \approx 1 + i\phi.$$

Therefore,

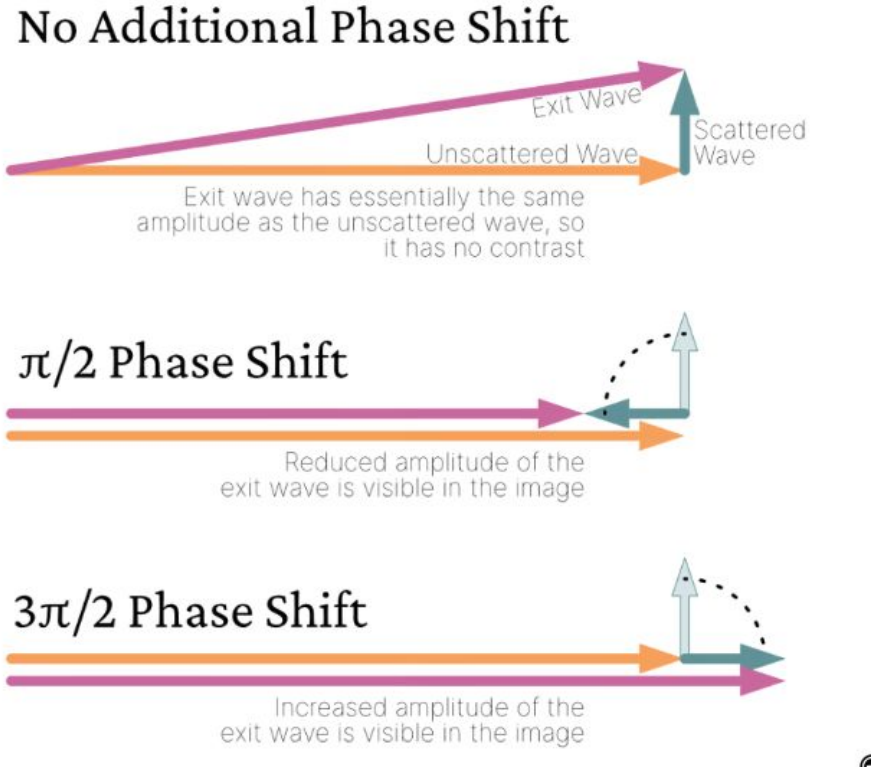
$$\Psi_{\text{exit}}(x, y) \approx \underbrace{1}_{\text{unscattered reference}} + \underbrace{i\phi(x, y)}_{\text{scattered wave}}$$

Because $i = e^{i\pi/2}$, this perturbation is in quadrature (90 degrees in phase) with the reference, when φ is real.

Also implies amplitude of exit wave after scattering at the protein remains at unity, so no contrast between unscattered reference at detector which only picks up intensity NOT the phase difference!

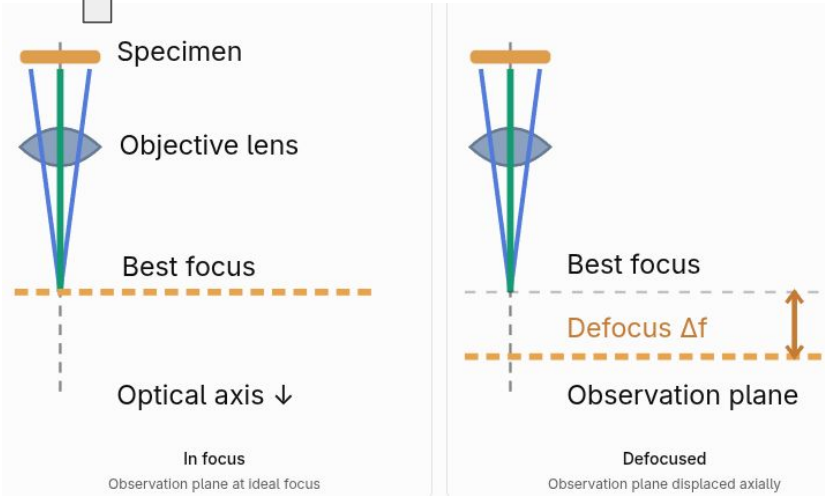
$$I(x, y) = |e^{i\varphi(x,y)}|^2 = 1.$$

Key principle of cryoEM:
Defocus changes the relative phase of the scattered Fourier components as they recombine with the unscattered reference.
Contrast: Image become visible when the scattered and unscattered wave constructively or destructively interact with each other which occurs when you add either pi/2 or 3pi/2 phase shift. So their intensity is no longer unity



Phasor diagram

Turn defocus knob to achieve this to get sample image

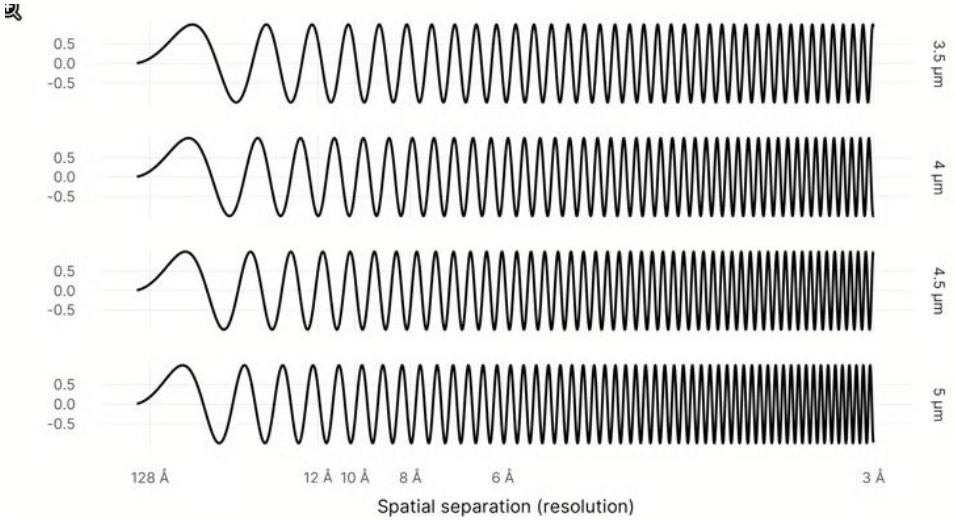


Each defocus value adds various contrast levels to different spatial resolutions

$$\mathrm{CTF}_{\mathrm{phase}}(q) = \sin \chi(q).$$

In practice images are taken and a large range of defocus values to ensure all spatial component contributions to the image can be seen i.e get their contrast

Where q is a given spatial frequency and chi can be seen as the relative phase shift(in radian) added a given spatial frequency contribution to the image



So plugging this into a sin function we see it gets max and min values at

$$x = \frac{\pi}{2} + 2\pi k$$

Which is precisely the phase shifts needed for each constructive and destructive interaction needed for maximum image contrast

Central slice theorem: each 2D Fourier image samples a central plane through the 3D Fourier volume.

$$(P_R V)(x, y) = \int_{-\infty}^{\infty} V\left(R^T (x, y, z)^T\right) \mathrm{d}z.$$

Integrates scattering potential V along beam direction z. R is rotation matrix so basically changing R changes the beam direction at which you take the image (see video 03, point 2: projection onto a detector)

The three-dimensional Fourier transform is

$$\widehat{V}(k_x, k_y, k_z) = \iiint V(x, y, z) e^{-2\pi i (k_x x + k_y y + k_z z)} \mathrm{d}x\mathrm{d}y\mathrm{d}z. \tag{10}$$

Now Fourier-transform the two-dimensional projection in Eq. (9):

$$\mathcal{F}_2\{PV\}(q_x, q_y) = \iint \left[\int V(x, y, z) \mathrm{d}z \right] e^{-2\pi i (q_x x + q_y y)} \mathrm{d}x\mathrm{d}y \tag{11}$$

$$= \iiint V(x, y, z) e^{-2\pi i (q_x x + q_y y + 0z)} \mathrm{d}x\mathrm{d}y\mathrm{d}z \tag{12}$$

$$= \boxed{\widehat{V}(q_x, q_y, 0).} \tag{13}$$

Therefore the 2D Fourier transform of a projection is a plane through the origin of the 3D Fourier transform. For an arbitrary orientation,

$$\boxed{\mathcal{F}_2\{P_R V\}(\boldsymbol{q}) = \widehat{V}\left(R^T (q_x, q_y, 0)^T\right).} \tag{14}$$

Forward Equation

Knowns
Y_i(projected image) can get directly from microscope reading
C_i can just calculate given a spatial resolution and defocus value

Reconstruction and refinement

$$Y_i(\mathbf{q}) = C_i(\mathbf{q})\widetilde{V}\left(R_i^\top \begin{bmatrix} q_x \\ q_y \\ 0 \end{bmatrix}\right) e^{-2\pi i \mathbf{q}\cdot\mathbf{t}_i} + N_i(\mathbf{q})$$

Unknowns
Pose(t,R)
V, 3D electron scattering potential

Putting everything together where Y_i is the fourier transform of the projected image, it can be modelled like this, where C_i is the CTF and t a real space translation(to correct frame shifts etc). A pose is (t,R) which has both orientation(R, beam direction) and translation t to reference plane. N is just the noise

If $T_{\mathbf{t}}g(\mathbf{x}) = g(\mathbf{x}-\mathbf{t})$, then

$$\mathcal{F}[T_{\mathbf{t}}g](\mathbf{k}) = e^{-2\pi i \mathbf{k}\cdot\mathbf{t}}\widehat{g}(\mathbf{k}). \tag{17}$$

A translation becomes multiplication by a unit-magnitude complex phase. If $h * g$ is a spatial convolution,

$$\mathcal{F}[h * g](\mathbf{k}) = \widehat{h}(\mathbf{k})\widehat{g}(\mathbf{k}). \tag{18}$$

ULTIMATE GOAL: obtain 3D electron potential in real space where we can fit our atomic model and get our final image

Reconstruction scheme 1: you know some poses

With poses held fixed, define A_i as the imaging operator (projection plus CTF plus shift). The measurement model is

$$\mathbf{y}_i = A_i \mathbf{v} + \boldsymbol{\varepsilon}_i.$$

Assume independent Gaussian pixel noise with covariance Σ_i. The negative log likelihood, up to constants, is

$$L(\mathbf{v}) = \frac{1}{2} \sum_{i=1}^{N} (\mathbf{y}_i - A_i \mathbf{v})^T \Sigma_i^{-1} (\mathbf{y}_i - A_i \mathbf{v}). \tag{16}$$

Taking a gradient with respect to $\mathbf{v}$ gives

$$\nabla L = \sum_i A_i^T \Sigma_i^{-1} (A_i \mathbf{v} - \mathbf{y}_i).$$

Setting this equal to zero yields the normal equations:

$$\boxed{\left(\sum_i A_i^T \Sigma_i^{-1} A_i \right) \widehat{\mathbf{v}} = \sum_i A_i^T \Sigma_i^{-1} \mathbf{y}_i.} \tag{17}$$

The right-hand side is a noise-weighted backprojection of the images. The left-hand matrix measures how strongly the data constrain different volume components. In practice these expressions are implemented using fast FFT, interpolation, and efficient GPU routines, not by forming the matrix explicitly.

Reconstruction scheme 2: you don't know any poses

reconstruction attempts to obtain a viable initial volume without assuming a known structure. CryoSPARC's original approach combines coarse-to-fine branch-and-bound pose search with stochastic gradient descent (SGD) on volume parameters [5, 15]. In SGD, a mini-batch $\mathcal{B}$ approximates the full dataset gradient:

$$V^{(t+1)} = V^{(t)} - \eta_t \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_V \ell_i(V^{(t)}).$$

The random batch reduces computational cost and introduces stochasticity, although it does not guarantee finding the global optimum.

Iterating a volume against noisy data creates a risk: noise can become reinforced by alignment to a reference that already contains the same noise. A standard defense is to split particles into two groups and independently refine two volumes V_A and V_B. Their reproducibility is tested using Fourier shell correlation (FSC):

$$\mathrm{FSC}(q) = \frac{\sum_{\mathbf{k} \in \mathcal{S}_q} \widehat{V}_A(\mathbf{k}) \widehat{V}_B(\mathbf{k})^*}{\sqrt{\sum_{\mathbf{k} \in \mathcal{S}_q} |\widehat{V}_A(\mathbf{k})|^2 \sum_{\mathbf{k} \in \mathcal{S}_q} |\widehat{V}_B(\mathbf{k})|^2}}. \tag{21}$$

Here $\mathcal{S}_q$ is a shell of Fourier frequencies near radius q; the reported FSC is ordinarily taken as the real shell